마이크로 스터러의 기하형상이 혈소판 활성화 및 응집에 미치는 영향

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Effect of micro-stirrer geometry on platelet activation and aggregation

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Key Words: Platelet(혈소판), Activation(활성화), Aggregation(응집)

Abstract

Platelets play a vital role in both hemostasis and thrombosis in blood circulatory systems. When platelets are exposed with high shear stress or with agonists, they are activated and aggregated. In our previous research, we had suggested that a new diagnostic method for platelet function assay using microfluidic system. In this study, we investigated effects of several micro-stirrers which had different contact areas to blood. The degree of platelet activation was estimated by calculating area of aggregated platelets in the microscope images. Shapes of stirrers vary from simple bar to rectangle. The experimental results yielded that platelet activation is strongly dependent on shearing time and contact area of stirrer. Time for full platelet activation decreased from 180sec to only 10sec with enlarged area of the stirrers. This dramatic decrease of the time for maximum activation of platelets could be useful for easy and rapid platelet function test.

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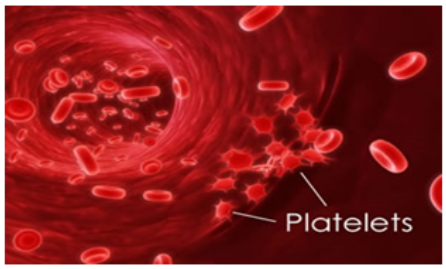
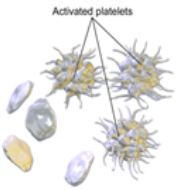
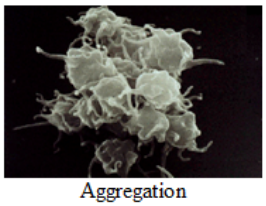
Shear Induced Platelet Activation (SIPA)

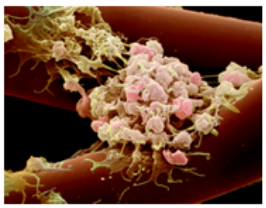




Platelet activation



Aggregation



Adhesion

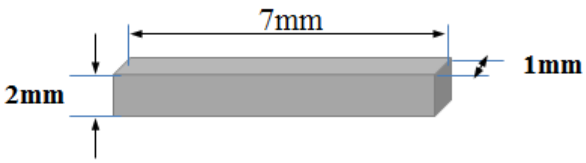
- **Platelet and its function for hemostasis**
  - One of the blood cells
    - Red blood cell, white blood cell and platelet
  - Biconvex discoid structure (≈ 2μm)
  - Platelet key function = hemostasis
  - Platelet activation
    1. By high shear
    2. By agonists (ADP or collagen)

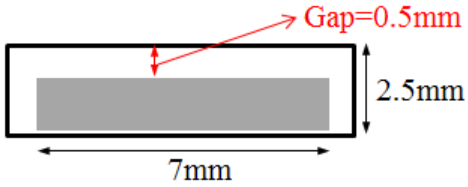
- **Shear-Induced Platelet Activation (SIPA)**
  - High shear stress result in below events on platelet.
    1. Activation
    2. Aggregation
    3. Adhesion
  - Shear stress is crucial for hemostasis and thrombosis at injured vascular regions.

Platelet activation according to stirrer geometries (contact area)

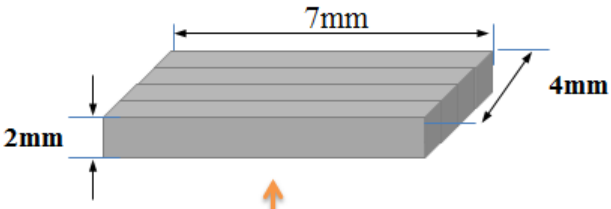
Change of a blood contact area

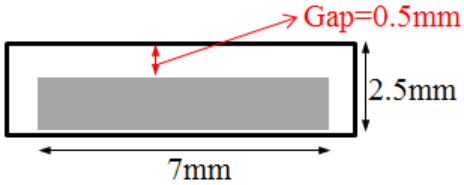
(A) Bar x 1





(B) Bar x 4

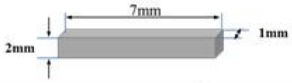


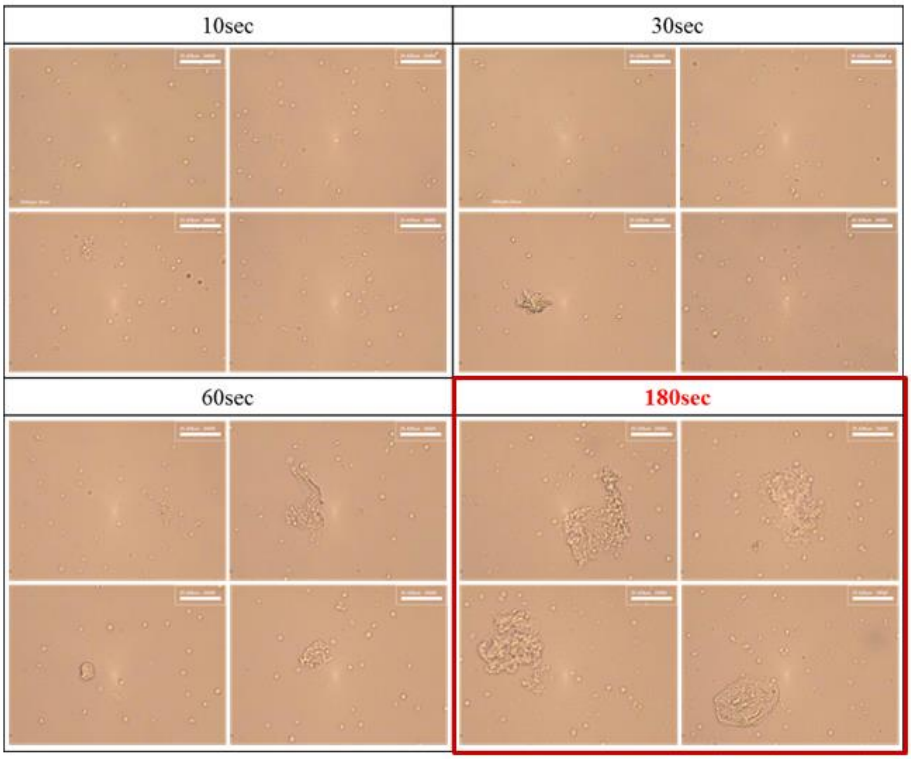


4 times Large contact area with blood

Platelet activation according to stirrer geometries (contact area)

(A) Bar x 1 for PRP at 2800rpm

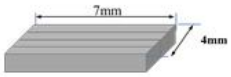


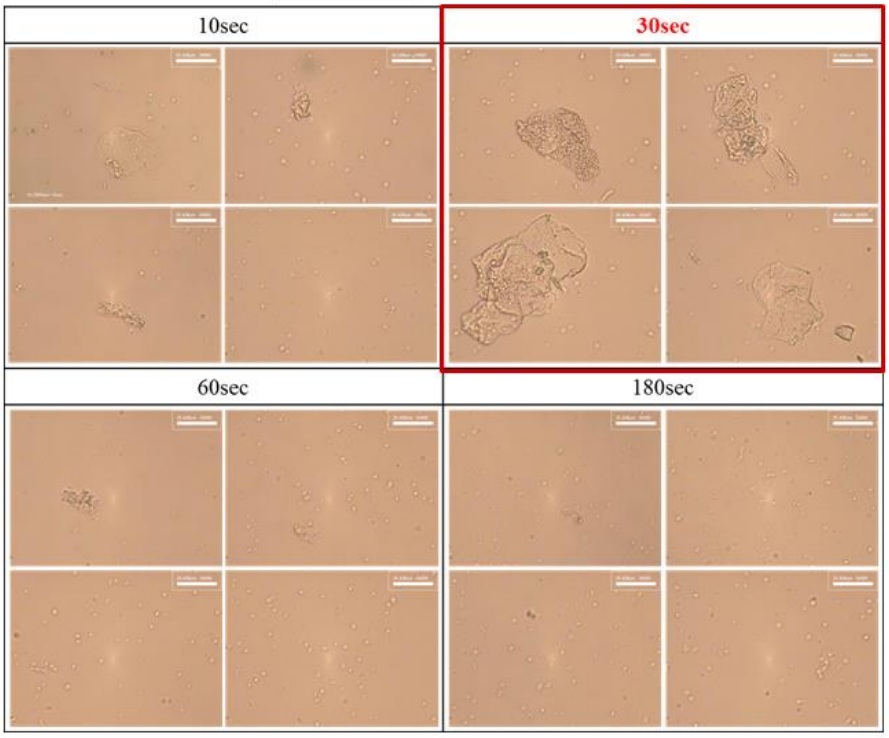


Maximized platelet activation

Platelet activation according to stirrer geometries (contact area)

(B) Bar x 4 for PRP at 2800rpm





Maximized platelet activation

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